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Assignment

Q1. What is the difference between AI and ML?

=> AI is the broad field focussed on creating systems that can perform tasks that can normally require human intelligence such as reasoning, understanding, decision-making and pattern recognition.

ML is a subset of AI that enables systems to learn from data and improve automatically without being explicitly programmed for every rule. ML focusses on creating algorithms that learn patterns from examples and make predictions or decisions based on data.

Q2. What do you mean by supervised or unsupervised learning?

=> Supervised learning is a type of machine learning in which the algorithm is trained on labelled

data - meaning each example has both input and correct output. The system learns to map input to output so that it can print output for new unseen input.

example -> Email spam detection - emails labelled 'spam' or 'not spam' and model learns to classify it.

Unsupervised learning uses unlabelled data where the correct outputs are not provided. Instead, the model tries to find patterns or structure in the data

example -> clustering customers into groups based on their shopping behavior without any predefined behaviors.

3. Explain reinforcement learning with example?

-> RL is a type of machine Learning where an agent learns by interacting with an environment and receiving feedback in the form of rewards or penalties. The agent explores different actions and learns to maximize cumulative reward over time through trial and error.

Example → A robot learning to navigate a maze.

- If the robot moves closer to goal it gets positive reward.
- If it hits a wall it gets negative reward.

4. How deep Learning is different from ML?

→ DL is a specialized subset of ML that uses multi-layered neural networks to automatically learn representations of data. deep means many layers in network that extracts hierarchical patterns in raw data.

Differences :-

- ML can use simple models (like SVMs, decision trees) that often require mannual feature engineering.
- DL uses neural networks that can automatically learn useful features from data especially for complex inputs such as images, speeches, texts, etc.

Q5. What is clustering? How it is different from classification

⇒ clustering is an unsupervised learning task where the algorithm groups data into clusters based on similarity without using predefined labels.

⇒ Classification is a supervised learning task where the model learns from labelled data.